

KKU-OIT iPBL Program 2026's Document

Joint Python Game Jam Workshop

Time: Sept. 8-11, 2026

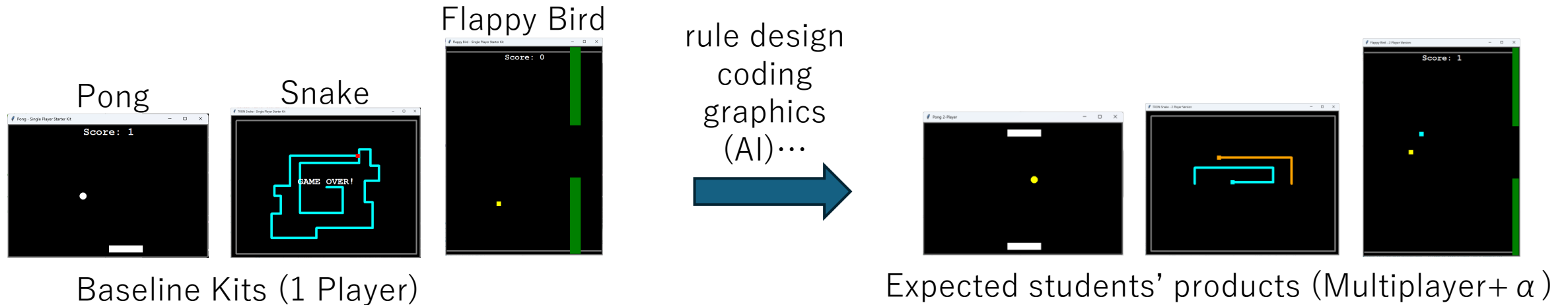
Place: College of Computing, Khon Kaen University, Thailand

3-Day International Game Jam Overview

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Topic:

Transform Single-Player Baselines into Creative Multiplayer Games



International Teamwork

- Mixed JP-TH Teams
- Cross-cultural peer learning

Inclusive Design

- Paper wireframe → Coding
- Open to all coding levels

Pure Python

- Standard **turtle** library
- Fantasy features (AI, ...) are welcomed

Sept. 8 (Tue) Day 1: Starter Kit Release, Paper Design & Core Coding

- AM 09:30-11:30
- Opening, team formation, **icebreak**, baseline release, rule explanation
 - Wireframe Design: draw 2P/4P concept on paper to map controls and assign roles.

- PM 13:00-16:30
- Start implement basic functions into a 2P/4P multiplayer game.

GOAL A functional single-player baseline with additional players rendered on screen.

Sept. 9 (Wed) Day 2: Multiplayer & Features

- AM 09:30-11:30
- 2P/4P collision physics, scoring system, winner/loser states.

- PM 13:00-16:30
- 2P/4P playtesting, advanced features (optional AI, Mediapipe/OpenCV).

GOAL A fully playable 2P/4P game with (optional) advanced features

Sept. 10 (Thu) Excursion

Sept. 11 (Fri) Day 3: Showcase & Tournament

- AM 09:30-11:30
- MOU Signing Ceremony
 - Refactor code (comments/clean loop) and prepare 3-min pitch slides.

- PM 13:00-16:30
- Pitching & arcade playtesting, evaluation, and award ceremony.

GOAL Final code submission, live team demonstration, and award presentation.

Day 1's Schedule

09:30-11:30

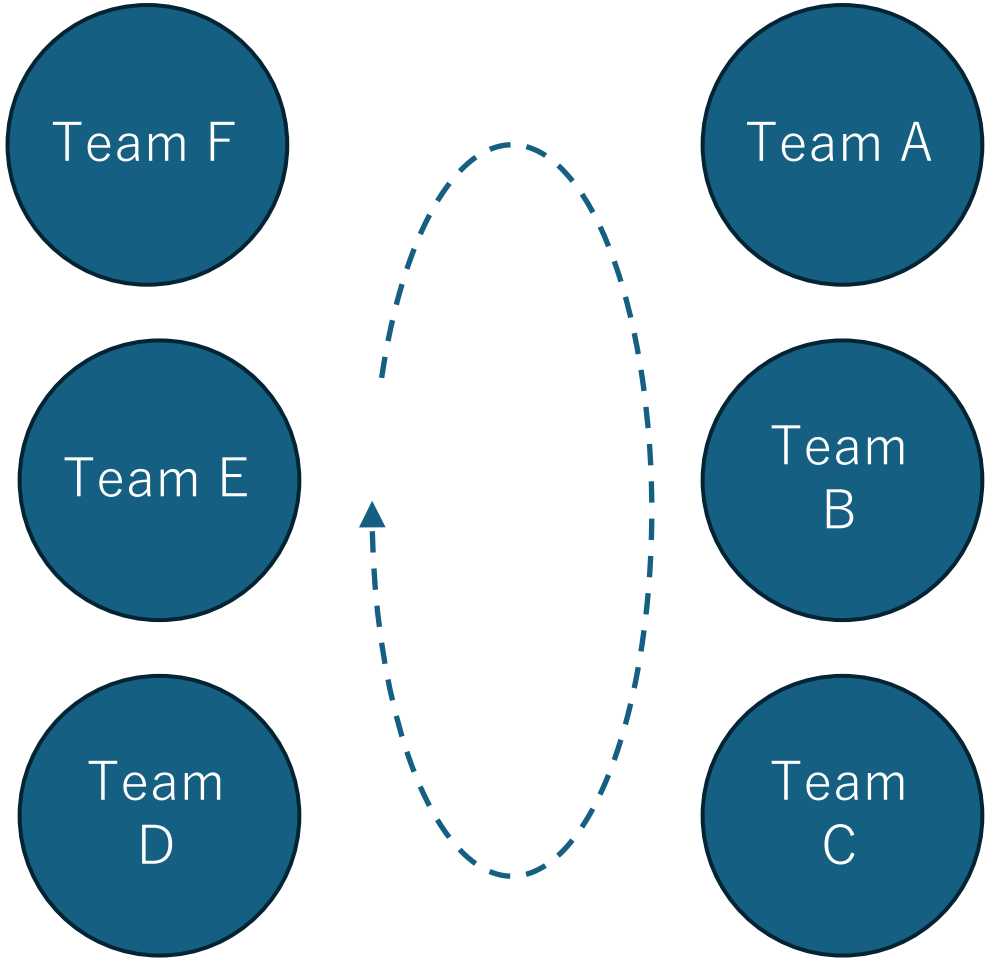
- Opening
- Team formation
- Rule explanation
- Baseline release, Python & IDE, Starter Kit check
- Wireframe design

13:00-16:30

- Implementation basic functions

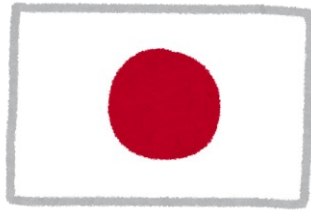
Seat locations

Screen/Balckboard



Icebreak

Introduce yourself to your teammates



Scope & Deliverables

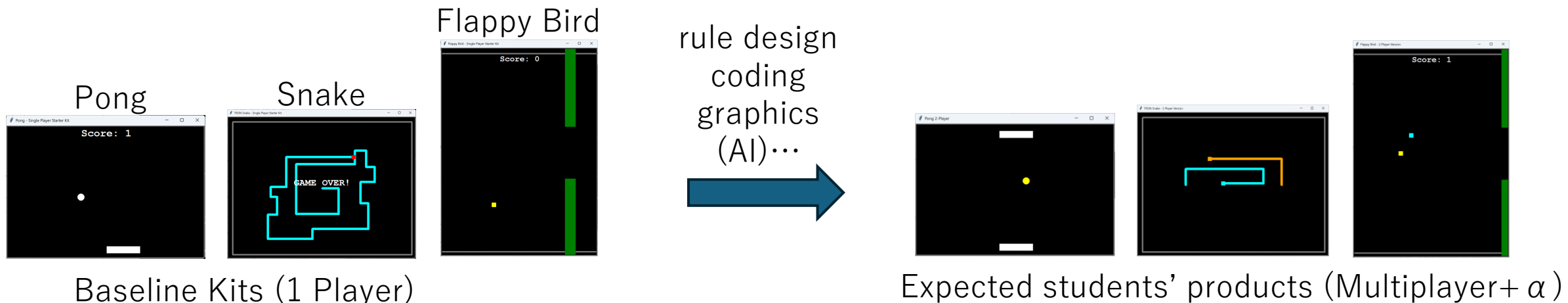
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Provided by Instructors:

- **Starter Kit:** Single-player baselines (Pong, TRON Snake, Flappy Bird) using Python turtle.
- **Hardware Support:** 1 USB keyboard per team for local multiplayer testing.

What You Must Build:

- **Phase 1 (Day 1):** Paper Wireframe (A3/A4 layout sketch submitted to mentors).
- **Phase 2 (Day 2 - Core):** Upgrade to 2P/4P Local Multiplayer on a single PC (WASD / Arrow Keys).
- **Phase 3 (Day 2 - Optional):** Advanced features (OpenCV, Mediapipe, AI Bot, ...).
- **Phase 4 (Day 3 - Final Demo):** Fully playable game + 3-minute pitch slide.



Development Rules & Guidelines

Workflow & Design Rules:

- Design First: Finish and submit your paper wireframe before writing code.
- Code Architecture: Follow the unified 5-section architecture in all baseline codebases.
- Code Understanding: Every team member must understand all lines of code (no copy-pasting blindly).
- Code Quality: Write clear Python comments throughout your code.

AI & Advanced Features Guidelines:

- Vibe Coding / AI Tools: Allowed only after submitting your paper wireframe.
- Advanced Features Policy: OpenCV, Mediapipe, or external libraries are welcome, but students must self-study; instructors will not provide technical support for these.

Development Rules & Guidelines

Clear Team Role Allocation (Recommended):

- **UI/UX & Visuals:** Design layouts, draw game assets, and handle UI feedback.
- **Core Game Logic:** Implement collision physics, scoring, and win/loss states.
- **Player Controls:** Handle multi-key input mapping (WASD, Arrow Keys, IJKL).
- **Special Features:** Manage optional mechanics (power-ups, AI, or OpenCV/Mediapipe).

Evaluation Criteria

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Evaluator	Criteria	Weight	What We Look For
Instructors (60%)	Code Architecture	20%	Modular structure, state management, clean logic.
	English Code & Docs	20%	100% English variable/function names and comments.
	Multiplayer Logic	20%	Smooth multi-player controls, stable FPS, accurate collision.
Peers (40%)	Fun & Playability	20%	Responsive controls, fair & engaging 2P/4P gameplay.
	Creative Features	20%	Power-ups, unique mechanics, visual polish.

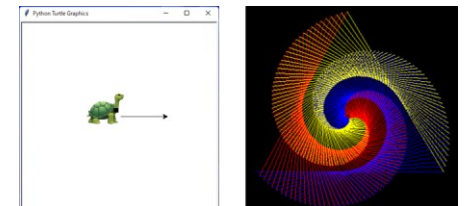
Baseline Release

- Pong: pong_baseline(1P).py
- TRON snake: snake_baseline(1P).py
- Flappy Bird: fbird_baseline(1P).py

- Python 3.14.3 + Turtle (Python's Standard Library)
- IDE (Integrated Development Environment) : VS Code or any other IDE
- If you want to install OpenCV/MediaPipe run command below on terminal of VS Code.

```
pip install opencv-python mediapipe
```

- About Turtle
 - Library to draw simple pictures.
 - Included in almost all python distributions.
 - ☞ Any PC with Python may run Turtle-based program without any problems!



Starter Kit Check

- Please plug in the USB drive provided by mentors to access the Starter Kit and Python environment.
- Run the files and play to check if Python and Turtle library are installed.
- Connect the USB keyboard and play the baselines to check if the keyboard works.

Pong Game

How to play?

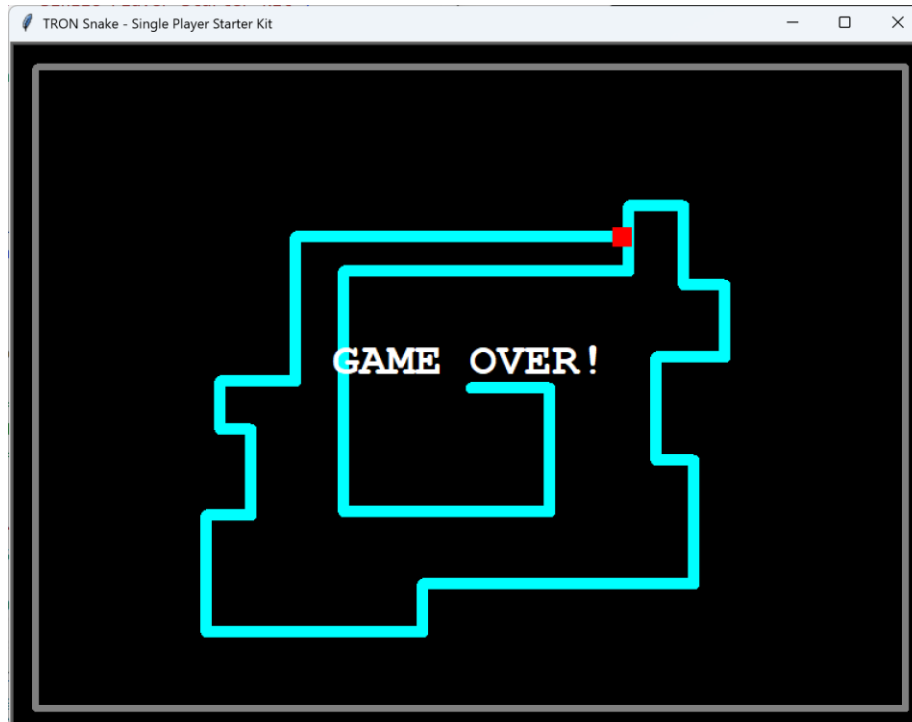
- Move the paddle by arrow keys ($\leftarrow$, $\rightarrow$)
- Bounce the ball so that it does not fall



TRON Snake Game

How to play?

- Move the snake's head by arrow keys ($\leftarrow$, $\rightarrow$, $\uparrow$, $\downarrow$)
- Keep the snake from hitting its body and the walls

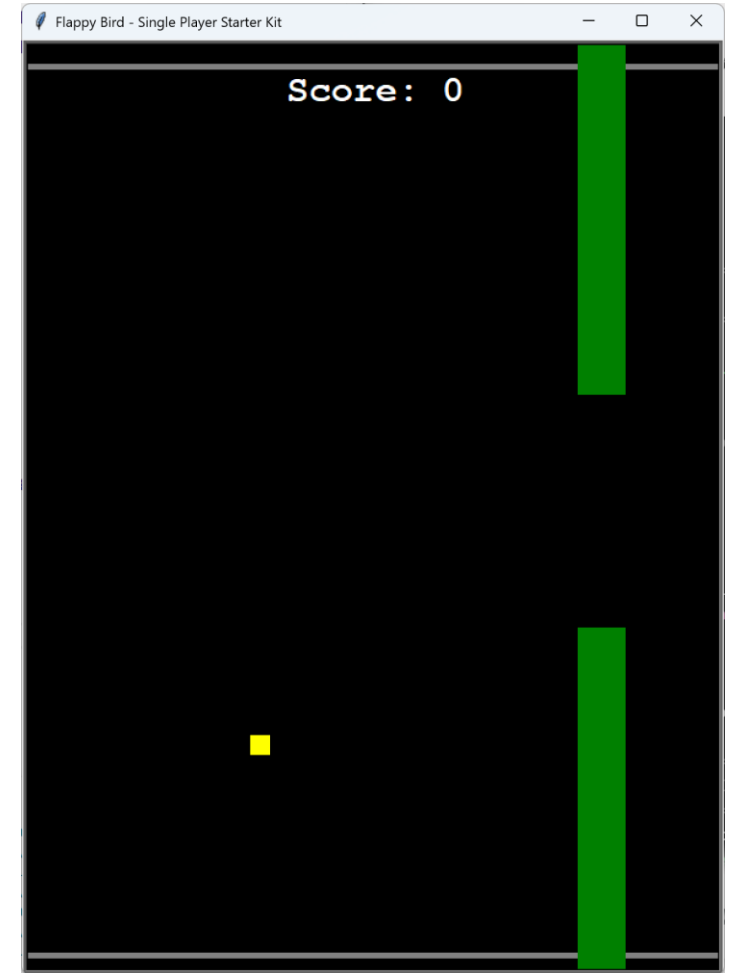


Flappy Bird Game

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How to play?

- Push Space bar to jump the bird so that the bird does not hit the pipes.



Hints for Creative Multiplayer Versions

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Game Mechanics & Mode Ideas:

- **Modes:** Free-for-all (2, 3, or 4 players) or Team vs Team (2v2 mode).
- **Dynamic Gameplay:** Add random/moving obstacles, environmental effects (wind, gravity), or power-ups to increase fun.
- **Scaling Difficulty:** Gradually increase game speed or mechanics complexity over time.

Visual & UI Balance:

- **Beautiful graphics** enhance user experience, but focus on core logic first—don't spend too much time polishing UI early on!

FORMAT OF BASELINE KITS

All games follow a standardized code structure to make your learning seamless:

- **Section 1: Screen Setup**

Configures window dimensions, title, background color, tracer settings...

- **Section 2: Game Entities Initialization**

- Creates and positions `turtle.Turtle()` objects (Ball, Pipes, Boundaries).
- Sets up the score display pen.

- **Section 3: Parameters & Physics**

Defines core physics variables (speed, direction, gravity, grid sizes).

- **Section 4: Input Handling**

Binds keyboard events (`wn.onkeypress`) to action functions.

- **Section 5: Main Game Loop**

Controls continuous frame updates using `wn.ontimer(game_loop, 16)` (about 60 FPS), physics calculations, collision checks...

Day 2's Schedule

Core Logic & Advanced Features

Focus Time — Make Your Game Fully Playable!

- Morning (09:30 – 11:30):
Collision Physics, Scoring System, Winner/Loser States.
- Afternoon (13:00 – 16:30):
Playtesting & Advanced Features (Optional: AI Bot, Mediapipe / OpenCV Controls).
- 🎯 End of Day Goal: A fully playable 2P/4P local multiplayer game.

Day 3's Schedule

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09:30-11:30

- MOU Signing Ceremony
- Refactor code (comments/clean loop)
- Prepare 3-min pitch slides.

13:00-16:30

- 13:00 – 13:30: Pitching Session (3 mins per team)
- 13:30 – 14:20: Arcade Playtesting & Evaluation Session
- 14:20 – 14:40: Break & Score Calculation
- 14:40 – 15:00: Instructor Feedback & Highlights
- 15:00 – 15:30: Closing & Award Ceremony
- 15:30 – 16:30: Photo Session & Venue Clean-up

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Awards & Certificates

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Grand Champion Award (Best Overall Game)

- **Highest total score** combined from Instructors (60%) and Peers (40%).



Technical Excellence Award

- **Highest Instructor score** for code architecture, English documentation, and logic execution (*excluding the Grand Champion winner*)



Best Gameplay Experience Award

- **Highest Peer evaluation score** for playability, fun factor, and creative mechanics. (*excluding previous award winners*)



Certificate of Participation

- Awarded to ALL students upon **completing the Game Jam showcase.**

Arcade Session Guide (1 of 2)

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1. Setup & Role Split

- Split your 4-member team into Group 1 (3 members) and Group 2 (3 members).
- Group 1: Hosts your booth first → Playtests second.
- Group 2: Playtests first → Hosts your booth second.

2. Fast-Paced Schedule (3 Mins / Booth)

- ROUND 1 (15 Mins) — Group 1 Playtests / Group 2 Hosts
 - Visit 5 opposing booths clockwise (2 mins play + 1 min form rating).
- 🔄 ROLE SWAP (2 Mins): Group 1 & 2 switch places!
- ROUND 2 (15 Mins) — Group 2 Playtests / Group 1 Hosts
 - Visit 5 opposing booths clockwise (2 mins play + 1 min form rating).

Arcade Session Guide (2 of 2)

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3. Host & Visitor Rules

- Hosts (At Home Booth): Run your game (`python abcxyz.py`), welcome visitors, and play with them!
- Visitors (Playtesters): Play for 2 minutes, scan the booth's QR Code, submit your score, and move immediately when the bell rings

4. Form for Peer Evaluation

<https://forms.gle/rjwDhVNuomAH3xUr7>

